ESTIMATOR-UPGRADE knob closure: dI-quadrature + amplitude-grid envelopes, a curvature-chord continuum no-condensate bound, and the controlled-error two-shell (0,0) Hessian

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The enumerated single-shell selection $\Delta F[\mathcal{R}] = F[\mathcal{R}] - F[\mathcal{R}_H] > 0$ ($\mathcal{R} \in$ {LAM, HEX, FCC, BCC} at $\mu^2 = 0.005$) was upgraded to a CONTROLLED-ERROR bound against the dominant M-quadrature knob in estimator-upgrade-enumerated v1.0 (curvature $\kappa_{\mathcal{R}} = \Delta F''_{\mathcal{R}}(0) \geq 0.85$, no condensate at $N_{\rm pt} = 6000$ and 20000). Two numerical knobs and the two-shell ensemble remained. This note closes the two residual single-shell knobs, upgrades the no-condensate verdict from a grid statement to a continuum one, and certifies the two-shell $(0,0)$ Hessian. Every free-energy value is produced by the certified m424 primitives (`dF_reading`, `dI`, `gap_solve`, `M_fast`); the knobs are exercised by monkeypatching m424 module globals, never by re-transcribing the free-energy algebra (code-discipline rule 1).

2. The two residual single-shell knobs and the two-shell question

The free-energy difference uses two quadratures beyond M: the loop log-difference `dI`$(\hat{r}, r_R)$ (a trapezoid on the grid $\_QG$ from `_grid_q`$(N_{\rm pt}, q_{\max})$) and the amplitude scan that decides $\min_{A>0} \Delta F$. The two-shell question is whether the $\{110\} + \{200\}$ ensemble of Math432 – in which the cross moment $m_{31} = \langle \phi_1^3 \phi_2 \rangle \neq 0$ makes the single-shell BCC point non-stationary – preserves the selection.

3. Method and results

(ii) amplitude-grid + continuum bound. For each reading the verdict $\min_{A>0} \Delta F > 0$ holds at $N_G = 121/241/481$ and ΔF is strictly increasing on the valid prefix (the unique minimum is $A = 0 = \mathcal{R}_H$). The "closest approach" $\min_{A>0} \Delta F$ itself scales as $\frac{1}{2}\kappa A_{\rm floor}^2 \to 0$ as the grid resolves near $A = 0$, so the grid-invariant is the binary verdict, not its magnitude. The node-free guarantee is the curvature-chord bound: a C^2 function deviates below its chord on $[A_i, A_{i+1}]$ by at most $\frac{1}{8} M_i \,\delta^2$ with M_i the local $|\Delta F''|$ from second differences, so $\Delta F(A) \geq \min(v_i, v_{i+1}) - \frac{1}{8} M_i \delta^2$.

reading	$\min_{A>0} \Delta F$ (N_G=481)	monotone	$M_{\max}$	continuum lower bound
LAM	1.17×10^{-6}	yes	126.0	$+8.80 \times 10^{-7}$
HEX	5.84×10^{-7}	yes	358.3	$+4.38 \times 10^{-7}$
FCC	4.76×10^{-7}	yes	526.8	$+3.57 \times 10^{-7}$
BCC	3.38×10^{-7}	yes	629.0	$+2.54 \times 10^{-7}$

The continuum lower bound is positive for every reading: no condensate can hide between grid nodes. Both the value ($\sim \frac{1}{2}\kappa A^2$) and the dip ($\sim \frac{1}{8}\kappa\delta^2$) scale as δ^2 near $A = 0$, and $\frac{1}{2} > \frac{1}{8}$, so the bound survives the near-origin region.

(ii) dI-quadrature knob. Refining _QG from $(6000, 50\,q_0)$ to $(12000, 100\,q_0)$ moves $\kappa_{\mathcal{R}}$ by $< 0.1\%$ and leaves every no-condensate verdict intact.

reading	κ (6000)	κ (12000)	$\lvert\Delta\kappa\rvert$
LAM	0.850966	0.850970	4.2×10^{-6}
HEX	2.554165	2.554177	1.3×10^{-5}
FCC	3.407767	3.407784	1.7×10^{-5}
BCC	5.116247	5.116273	2.5×10^{-5}

Combined with the M-knob envelope of estimator-upgrade-enumerated v1.0, the single-shell margins are controlled with respect to all three numerical knobs (M, `dI`, amplitude grid).

(i) two-shell $(0, 0)$ **Hessian.** For the $\{110\} + \{200\}$ condensate (A_1, A_2) the quadratic cross term vanishes: the shells are disjoint, so $\langle\phi_1\phi_2\rangle = 0$ and the m_{31} coupling $U\langle\phi_1^3\phi_2\rangle$ is QUARTIC ($A_1^3A_2$), absent from the Hessian. The Hessian is therefore diagonal, $\kappa_{12} = 0$. The soft eigenvalue is the single-shell BCC curvature $\kappa_{\rm BCC} = 5.116$ (controlled-error; the two-shell F at $A_2 = 0$ is exactly the single-shell BCC F). The $\{200\}$ eigenvalue is strictly larger: that shell sits one octave off the kernel minimum and carries a positive penalty $Cq_0^4 = 0.214 > 0$. Hence $(0, 0)$ is a strict two-shell minimum with controlled error. The GLOBAL two-shell no-condensate (the m_{31} tilt valley) is the exact-Wick PASS of Math432 (anchored $\Delta F > 0$ on the (A_1, A_2, M) surface at cut$12/48^3$ and cut$20/64^3$), cited here as grid evidence and not re-derived.

4. What this advances and what remains

ADVANCED: the single-shell enumerated $\Delta F > 0$ verdict is now controlled-error against all three numerical knobs and the no-condensate statement is a continuum (node-free) one; the two-shell $(0, 0)$ Hessian is positive-definite with controlled error. REMAINING for full ESTIMATOR-UPGRADE closure: the two-shell GLOBAL no-condensate is still grid evidence (Math432 exact-Wick scan), not a continuum bound; a curvature-chord or interval bound over the two-shell (A_1, A_2) surface would close it. The gate stays OPEN pending that and operator sign-off.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) *"The curvature-chord M_i is a discrete estimate of $\lvert\Delta F''\rvert$, not a proven upper bound."* VALID-with-mitigation: M_i is the local second-difference, an $O(\delta^2)$-accurate estimate; the bound is therefore controlled-error / STRONG-EVIDENCE grade, not a T7 theorem. Mitigation: M_i is grid-stable (the second differences converge as $\delta \to 0$) and the bound clears zero by a factor ~ 3 near the origin, so a modest M under-estimate cannot flip the sign. Recorded as STRONG-EVIDENCE, not as a proof.

(β) *"The two-shell global no-condensate is only cited, not re-derived – the gate cannot close on a citation."* UPHELD: correct, and it is exactly why this note does NOT claim ESTIMATOR-UPGRADE closure. The two-shell $(0, 0)$ Hessian is new controlled-error content; the global valley remains Math432 grid evidence. The residual is named in section 4 and the gate stays OPEN.

(γ) *"Is the soft two-shell eigenvalue really the single-shell BCC curvature?"* DISMISSED: the two-shell free energy restricted to $A_2 = 0$ is identically the single-shell BCC free energy (Math432 reproduces the Math430 single-shell race at $A_2 = 0$ to 10^{-7}), so the A_1 curvature at $(0, 0)$ is κ_{BCC} exactly; the script reproduces 5.116 to 5×10^{-3}.

Quantitative sanity check (mandatory): magnitude – κ envelopes $\sim 10^{-5}$ vs $\kappa \sim$ 1–5 ($< 0.1\%$); the continuum dip $\frac{1}{8}M\delta^2 \sim 10^{-7}$ vs value $\sim 10^{-6}$; sign-direction – every $\kappa > 0$ and every continuum bound > 0 (the selection-preserving direction); limit-case – $\Delta F(0) = 0$ exactly; consistency – the $\{200\}$ penalty $Cq_0^4 = 0.214 > 0$ confirms the stiff shell adds curvature, so the soft direction $\kappa_{\mathrm{BCC}} = 5.116$ is the binding two-shell eigenvalue.

6. Result footer

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Result ID:               B1-RH-ENUM / ESTIMATOR-UPGRADE knob closure (single-shell)
Precise statement:       For LAM/HEX/FCC/BCC at mu^2=0.005: (ii) kappa_R moves
                         < 0.1% under dI-grid refinement (6000,50)->(12000,100)
                         and the no-condensate verdict is grid-monotone at
                         NG=121/241/481; (iii) the curvature-chord continuum
                         lower bound min(v_i,v_{i+1}) - (1/8) M_i delta^2 > 0 on
                         every A>0 interval (no inter-node condensate); (i) the
                         two-shell {110}+{200} (0,0) Hessian is positive-definite
                         (kappa_12=0 by orthogonality; soft eigenvalue kappa_BCC
                         =5.116; {200} penalty C q0^4=0.214>0). Global two-shell
                         no-condensate cited from Math432 (grid evidence).
Scope:                   Single-shell ensemble, all 3 numerical knobs + continuum
                         no-condensate; two-shell (0,0) curvature only.
Dependencies:            Math424_AddA_reading_uniqueness (dF_reading, dI,
                         gap_solve, M_fast); estimator-upgrade-enumerated v1.0
                         (M-knob); Math432 (two-shell global no-condensate).
Evidence grade:          EXECUTED (estimator_upgrade_knobs.py 13/13) +
                         STRONG-EVIDENCE (curvature-chord continuum bound).
Reproduction command:    python codes/vacuum/estimator_upgrade_knobs.py
Expected output:         13/13 PASS; continuum lower bounds positive; dI
                         envelope < 0.1%; two-shell Hessian PD.
Falsification gate:      A reading with a negative continuum lower bound; a dI
                         envelope > 0.1% of kappa; a two-shell direction with
                         non-positive (0,0) curvature; or a Math432 reload with
                         an anchored dF < 0.
Tier before / after:     No tier/gate flip. ESTIMATOR-UPGRADE: single-shell knob
                         closure + continuum no-condensate + two-shell (0,0)
                         curvature ADVANCE; gate stays OPEN pending the two-shell
                         global continuum bound + operator sign-off.
No-overclaim statement:  Controlled error w.r.t. the named numerical knobs and the
                         two-shell (0,0) curvature only. NOT a full ESTIMATOR-
                         UPGRADE closure: the two-shell global no-condensate is
                         cited Math432 grid evidence, not a continuum bound. The
                         B1 T6 selection SIGN was already established; this
                         upgrades the enumerated MARGINS' error grade.
Next required action:    Two-shell global continuum no-condensate bound (curvature
                         -chord over the (A1,A2) surface); then ESTIMATOR-UPGRADE
                         closure is an operator decision.
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